1. Find the partial derivatives of $f(s,t) = \dfrac{s-t}{s+t}$.

2. Find the partial derivatives of $f(x,y) = \displaystyle\int_x^{y^2} e^{t^2}\,dt$.

3. Show that the heat equation
$$\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2}$$
is satisfied by $u(x,t) = 10e^{-t}\sin x$.

4. Evaluate the limit
$$\lim_{(x,y)\to(0,1)} \frac{y \sin x}{x(y+1)}.$$

5. Evaluate the limit
$$\lim_{(x,y)\to(0,0)} \frac{|xy|}{xy}.$$

6. Evaluate the limit
$$\lim_{(x,y)\to(1,0)} \frac{\sin xy}{xy}.$$

7. Use the epsilon-delta definition of the multivariable limit to show: If

$$\lim_{t \to a} f(t) = L$$

and

$$\lim_{(x,y) \to (b,c)} g(x,y) = a$$

then

$$\lim_{(x,y) \to (b,c)} f(g(x,y)) = L.$$

8. Verify that $\lim_{(x,y) \to (0,0)} \dfrac{\sin x + \sin y}{x + y} = 1.$